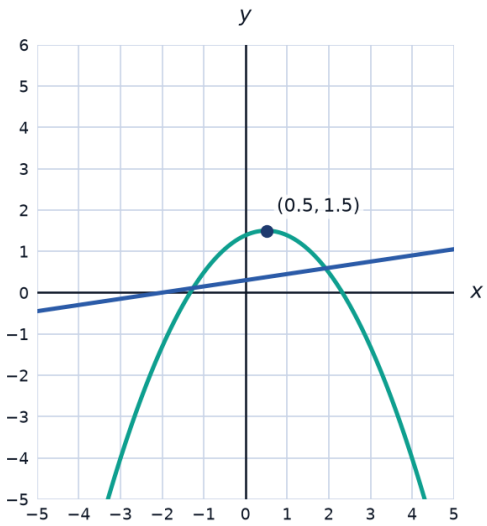
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int x^4\,dx$.

**2** **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int \sqrt{x}\,dx$.




**3** **[Foundational]** • **Target: Basic antiderivatives (the power rule)**

Find $\int \frac{1}{x^3}\,dx$.

**4** **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (4x^3 - 6x + 5)\,dx$.

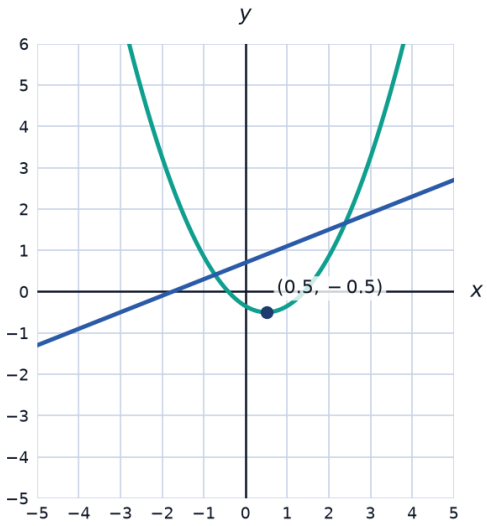
**5** **[Foundational]** • **Target: Antiderivatives of polynomials**

Find $\int (2x + 7)\,dx$.

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**6** [Foundational] • Target: Antiderivatives of polynomials

Find $\int (x^2 - 4x + 3)\,dx$.




**7** [Foundational] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \cos x\,dx$.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \sin x\,dx$.

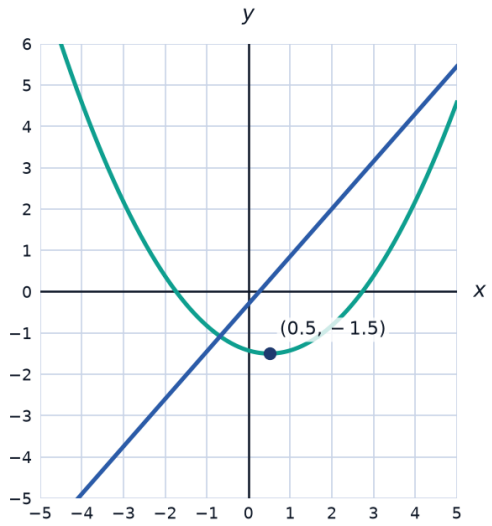
**9** [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int e^x\,dx$.

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**10** [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_0^2 3x^2\,dx$.




**11** [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_1^3 2x\,dx$.

**12** [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_{-1}^2 3x^2\,dx$.

**13** [Intermediate] • Target: Definite integrals with polynomial integrands

Evaluate $\int_0^2 (x^2+1)\,dx$.

**14** [Intermediate] • Target: Definite integrals with polynomial integrands

Evaluate $\int_1^3 (2x^2-3x+1)\,dx$.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

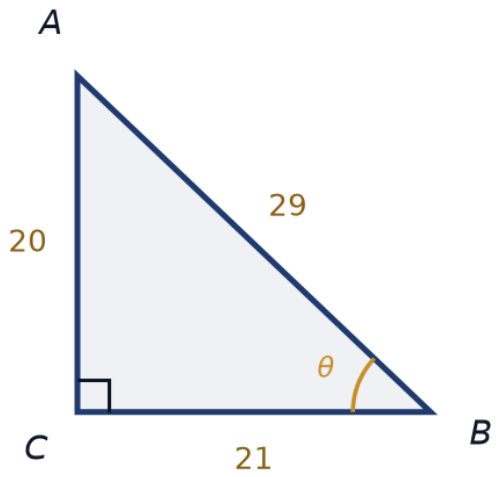
**15** [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\int_{-2}^1 (x^2-x)\,dx$.

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**16** [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 6.01 Limits & Derivatives

Evaluate $\int_0^{\pi/2} \cos x \, dx$.




**17** [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

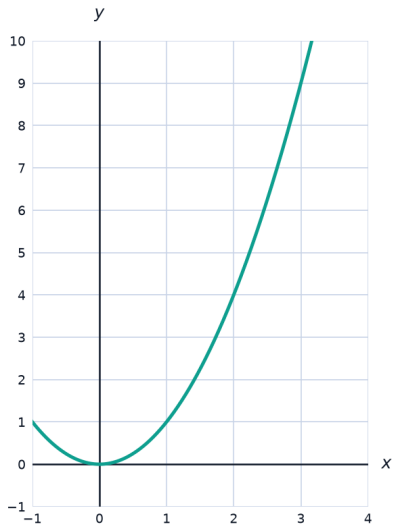
Evaluate $\int_0^{\pi} \sin x \, dx$.

**18** [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\int_0^{1} e^x \, dx$.

**19** [Challenging] • Target: Area under a curve • Cross-Lesson: 1.01 Algebra & Equations

Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.




**20** [Challenging] • Target: Area under a curve • Cross-Lesson: 2.01 Functions & Transformations

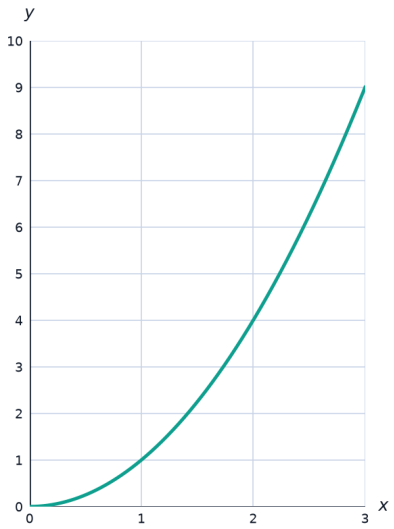
Find the area under $y = 4 - x^2$ from $x = -2$ to $x = 2$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** **[Foundational]** • **Target: Area under a curve**

The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.




**a.** 9

**b.** 18

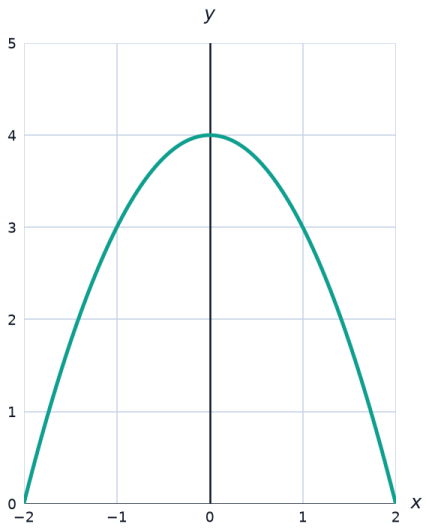
**c.** 3

**d.** 27

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**22** [Foundational] • Target: Area under a curve

The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.




**a.** $\frac{16}{3}$

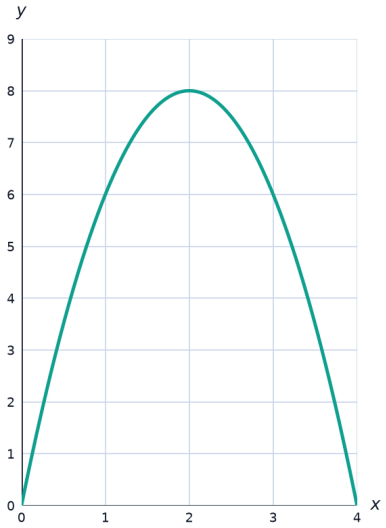
**b.** 32

**c.** 16

**d.** $\frac{32}{3}$

**23** [Foundational] • Target: Word problems: integration in context

The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \le t \le 4$. Find the particle's displacement (the area under the curve).




**a.** $\frac{64}{3}$

**b.** 16

**c.** 32

**d.** 64

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**24** [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int \sqrt[3]{x}\, dx$.

**a.** $\frac{3}{2}x^{2/3} + C$

**b.** $\frac{4}{3}x^{4/3} + C$

**c.** $\frac{3}{4}x^{4/3} + C$

**d.** $\frac{1}{3}x^{-2/3} + C$

**25** [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int 7\, dx$.

**a.** $C$

**b.** $7x + C$

**c.** $7 + C$

**d.** $\frac{7x^2}{2} + C$

**26** [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int \frac{1}{\sqrt{x}}\, dx$.

**a.** $2\sqrt{x} + C$

**b.** $\frac{1}{2}\sqrt{x} + C$

**c.** $-2\sqrt{x} + C$

**d.** $\sqrt{x} + C$

**27** [Foundational] • Target: Antiderivatives of polynomials

Find $\int (3x^3 - 2x)\, dx$.

**a.** $3x^4 - x^2 + C$

**b.** $\frac{3x^4}{4} - x^2 + C$

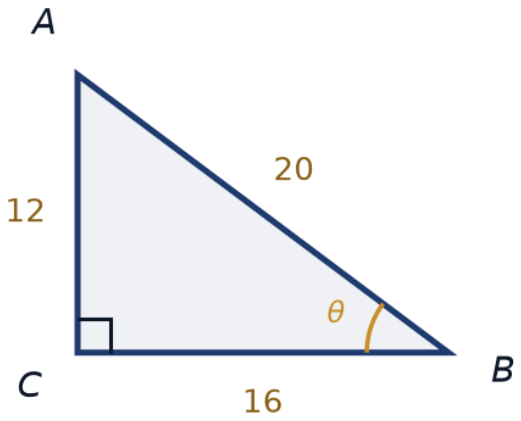
**c.** $\frac{3x^4}{4} - 2x^2 + C$

**d.** $\frac{x^4}{4} - x^2 + C$

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**28** [Foundational] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int 2\cos x\,dx$.




**a.** $2\sin x + C$

**b.** $-2\sin x + C$

**c.** $2\cos x + C$

**d.** $\sin(2x) + C$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int 3\sin x\,dx$.

**a.** $3\sin x + C$

**b.** $-3\sin x + C$

**c.** $3\cos x + C$

**d.** $-3\cos x + C$

**30** [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int (e^x + 2)\,dx$.

**a.** $e^x + 2 + C$

**b.** $e^x + 2x + C$

**c.** $xe^x + 2x + C$

**d.** $e^x + C$

**31** [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int (\sin x + \cos x)\,dx$.

**a.** $-\cos x - \sin x + C$

**b.** $\cos x + \sin x + C$

**c.** $\cos x - \sin x + C$

**d.** $-\cos x + \sin x + C$

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**32** [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_2^5 3\,dx$.

**a.** 3

**b.** 9

**c.** 6

**d.** 15

**33** [Intermediate] • Target: Definite integrals of trigonometric and exponential functions

Evaluate $\int_0^{\pi/2} \sin x\,dx$.

**a.** $\frac{\pi}{2}$

**b.** $-1$

**c.** 0

**d.** 1

**34** [Intermediate] • Target: Definite integrals of trigonometric and exponential functions

Evaluate $\int_0^{\pi} \cos x\,dx$.

**a.** $\pi$

**b.** $-2$

**c.** 0

**d.** 2

**35** [Intermediate] • Target: Signed area and regions below the axis

Evaluate $\int_0^4 (x-4)\,dx$ and state whether the region is above or below the $x$-axis.

**a.** 8; above

**b.** $-8$; above

**c.** 8; below

**d.** $-8$; below

**36** [Intermediate] • Target: Properties of definite integrals

Given $\int_1^5 f(x)\,dx = 20$ and $\int_1^5 3\,dx = 12$, find $\int_1^5 [f(x) - 3]\,dx$.

**a.** $-8$

**b.** 8

**c.** 32

**d.** 17

**37** [Intermediate] • Target: Properties of definite integrals

Given $\int_0^4 f(x)\,dx = 9$ and $\int_0^4 g(x)\,dx = 5$, find $\int_0^4 [f(x) + g(x)]\,dx$.

**a.** $\frac{9}{5}$

**b.** 45

**c.** 4

**d.** 14

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Finding a function from its derivative • Cross-Lesson: 9.01 Statistics & Probability

A function satisfies $f'(x) = 2x + 7$ and $f(-1) = 4$. Find $f(x)$.

**a.** $x^2 + 7x + 10$

**b.** $2x^2 + 7x + 10$

**c.** $x^2 + 7x - 10$

**d.** $x^2 + 7x + 4$

**39** [Challenging] • Target: Average value of a function • Cross-Lesson: 1.01 Algebra & Equations

Find the average value of $f(x) = x^2 - 1$ on $[1, 3]$.

**a.** $\frac{8}{3}$

**b.** $\frac{10}{3}$

**c.** $\frac{20}{3}$

**d.** $6$

**40** [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 2.01 Functions & Transformations

A particle moves along a line with velocity $v(t) = 8t - 2t^2$ (m/s) for $0 \le t \le 4$. Find the particle's displacement over this interval.

**a.** $64$

**b.** $\frac{64}{3}$

**c.** $32$

**d.** $\frac{192}{3}$

**41** [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 3.01 Sequences & Series

Water flows into a tank at a rate of $r(t) = 30 - 3t$ liters per minute, for $0 \le t \le 10$. Find the total volume of water that flows in during this time.

**a.** $300$

**b.** $150$

**c.** $450$

**d.** $100$

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**42** [Challenging] • Target: Word problems: integration in context • Cross-Lesson: 4.01 Trigonometry

A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \le t \le 8$. Find the total number of items produced during this 8-hour period.

**a.** 528

**b.** 656

**c.** 450

**d.** 400

**43** [Challenging] • Target: Integration & Applications • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Find $\int -x^4\,dx$.

**a.** $-\frac{1}{5}x^5 + C$

**b.** $-\frac{1}{4}x^4 + C$

**c.** $-x^5 + C$

**d.** $-\frac{1}{6}x^5 + C$

**44** [Challenging] • Target: Integration & Applications • Cross-Lesson: 6.01 Limits & Derivatives

Find $\int -3x^4\,dx$.

**a.** $-3x^5 + C$

**b.** $-\frac{3}{5}x^5 + C$

**c.** $-\frac{1}{2}x^5 + C$

**d.** $-\frac{3}{4}x^4 + C$

**45** [Challenging] • Target: Integration & Applications • Cross-Lesson: 8.01 Exponentials & Logarithms

In a laboratory calibration sequence, find $\int (-2x + 5)\,dx$.

**a.** $-2x + 5 + C$

**b.** $-x^2 + 7x + C$

**c.** $-x^2 + 5x + C$

**d.** $-x^2 - 5x + C$

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